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REVIEW

Artificial Intelligence Application in Diagnosing, Classifying, Localizing, Detecting and Estimation the Severity of Skin Condition in Aesthetic Medicine: A Review

Alvin Kar Wai Lee[1] | Lisa Kwin Wah Chan[1] | Cheuk Hung Lee[1] | Jair Mauricio Cerón Bohórquez[2] | Diala Haykal[3] | Jovian Wan[4] | Kyu-Ho Yi[5,6]

[1]EverKeen Medical Centre, Hong Kong, China | [2]Medical Contour Hamburg, Hamburg, Germany | [3]Centre Laser Palaiseau, Palaiseau, France | [4]Medical Research Inc., Wonju, South Korea | [5]Division in Anatomy and Developmental Biology, Department of Oral Biology, Human Identification Research Institute, BK21 FOUR Project, Yonsei University College of Dentistry, Seoul, Korea | [6]Maylin Clinic (Apgujeong), Seoul, Korea

Correspondence: Kyu-Ho Yi (kyuho90@daum.net)

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ABSTRACT

Background: The integration of artificial intelligence (AI) and machine learning (ML) has revolutionized aesthetic medicine, enhancing the diagnosis, classification, and treatment of skin conditions. These technologies offer high precision, personalized care, and the potential to reduce human error. This review aimed to evaluate the current applications of AI and ML in aesthetic medicine, focusing on studies graded as Level I or II evidence by the Oxford Centre for Evidence-Based Medicine (CEBM).

Methods: A comprehensive search of MEDLINE, PubMed, and Ovid databases identified studies employing AI and ML for diagnosing and managing skin conditions. Studies were included if they demonstrated high diagnostic accuracy, improved treatment personalization, or other measurable clinical outcomes.

Results: AI and ML systems showed high accuracy in detecting and diagnosing conditions such as skin cancer, acne, psoriasis, and seborrheic dermatitis. AI-based platforms facilitated personalized treatment plans, enhancing therapeutic outcomes while minimizing errors. The integration of AI reduced diagnostic time and lowered healthcare costs, demonstrating significant potential for improving patient care. However, challenges such as algorithmic bias, data privacy concerns, and the need for high-quality training datasets were highlighted.

Conclusion: AI and ML have transformative potential in aesthetic medicine, offering improved diagnostic precision, enhanced patient outcomes, and cost reductions. Addressing limitations related to algorithm bias, regulatory oversight, and data quality is essential to fully realize the benefits of AI in clinical practice. Future research should focus on developing robust, ethical, and regulatory-compliant AI solutions.

1 | Introduction

The field of aesthetic medicine has witnessed a significant transformation in recent years, with the integration of artificial intelligence (AI) and machine learning (ML) playing a crucial role in revolutionizing the way dermatologists and healthcare professionals diagnose, treat, and manage various skin conditions. The rapid advancements in AI and ML have enabled the development of sophisticated algorithms that can analyze large datasets, identify patterns, and make predictions with high

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accuracy. This has led to a new era of precision medicine, where AI-powered systems can provide personalized treatment plans and improve patient outcomes [1, 2].

Aesthetic medicine encompasses a broad range of skin conditions, including skin cancer, acne, psoriasis, seborrheic dermatitis, and others. These conditions can significantly impact a patient's quality of life, leading to emotional distress, social isolation, and reduced self-esteem. Conventional diagnostic methods often rely on visual examination and clinical judgment, which can be subjective and prone to error. In contrast, AI-powered systems can analyze vast amounts of data, including images, text reports, and patient history, to provide accurate diagnoses and treatment recommendations [3, 4].

The integration of AI and ML in aesthetic medicine is not limited to diagnosis alone. AI-powered systems can also assist in treatment planning, monitoring treatment response, and predicting outcomes. For instance, AI-powered systems can analyze patient images to detect early signs of skin cancer, providing clinicians with valuable insights to develop targeted treatment plans. Similarly, AI-powered systems can monitor treatment response in real-time, enabling clinicians to adjust treatment protocols as needed [5].

The benefits of AI-powered systems in aesthetic medicine are multifaceted. Firstly, they can improve diagnostic accuracy by reducing human error and bias. Secondly, they can enhance patient outcomes by providing personalized treatment plans tailored to individual needs. Thirdly, they can reduce healthcare costs by minimizing unnecessary procedures and treatments. Lastly, they can improve patient engagement by providing them with real-time information and empowering them to take a more active role in their care [6, 7].

This review aims to provide a comprehensive overview of the current state of AI-powered systems in aesthetic medicine, highlighting their benefits and limitations. The review will focus on studies that have been graded as Level I or II evidence by the Oxford Centre for Evidence-Based Medicine (CEBM) [8], providing a robust foundation for understanding the role of AI-powered systems in this field. By exploring the current landscape of AI-powered systems in aesthetic medicine, this review aims to inform clinicians about the potential benefits and challenges associated with these technologies and provide guidance on their implementation in clinical practice.

2 | Materials and Methods

Keywords including “Artificial Intelligence”, “Machine Learning,” “Deep Learning,” “Aesthetic Medicine,” “Cosmetic,” “Cosmetic Dermatology,” “Cosmetic Medicine,” were searched in the MEDLINE, PubMed and Ovid databases for relevant studies published on photoaging treatment. Some papers were further reviewed using a double-blinding approach, sample size, control usage, randomization usage and objective endpoint measurements. All studies were classified according to the Oxford Center for evidence-based medicine evidence hierarchy.

3 | Results

3.1 | Classification of Skin Conditions

The results are summarized in Table 1.

Molton et al. [21] presents a novel system for automatic three-dimensional (3D) analysis of facial rejuvenation. The system aims to provide a more accurate and objective assessment of facial aging and rejuvenation, which is essential for cosmetic medicine.

The authors developed a 3D analysis system that uses a combination of computer vision and ML algorithms to analyze facial images and generate a 3D model of the face. The system evaluates various facial features, including skin texture, wrinkles, and facial asymmetry, to provide a comprehensive assessment of facial rejuvenation.

The system was tested on a dataset of 30 patients who underwent facial rejuvenation surgery, with impressive results. The system demonstrated high accuracy in detecting changes in facial features after surgery, with an average accuracy of 92%. The authors also compared the results with manual assessments by experts and found that the system was more consistent and objective (Level IIb).

Jeong et al. [22] presents the development of an AI system called AI-ScalpGrader for diagnosing and classifying scalp health. The authors designed a deep learning (DL)-based AI system using convolutional neural networks (CNNs) to analyze images of scalp conditions, including seborrheic dermatitis, psoriasis, and healthy scalp.

The AI-ScalpGrader system was trained using a dataset of 1200 images of scalp conditions and tested on a separate dataset of 300 images. The results showed that the AI system achieved high accuracy in diagnosing and classifying scalp conditions, with an overall accuracy of 94.3%. The system was also able to detect subtle differences between conditions, such as the presence of inflammation or crusts.

The authors compared the performance of the AI-ScalpGrader system with that of human dermatologists and found that the system outperformed human experts in terms of accuracy and consistency. The system also demonstrated its ability to learn from new data and adapt to changes in scalp conditions over time (Level IV).

Kim et al. [23] presents a novel approach to facial skin problem diagnosis using a combination of computer vision and DL techniques. The authors propose an enhanced Mask R-CNN (Masked Region-based Convolutional Neural Networks) architecture, which is trained on a large dataset of facial images with annotated skin problems, such as acne, dark spots, and fine lines.

The proposed system uses a two-stage approach: first, the Mask R-CNN detects the region of interest (ROI) in the facial image, and then a super-resolution Generative Adversarial Network (SRGAN) is used to enhance the resolution of the ROI and

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TABLE 1 | The table summarized each study on skin condition detection and localization.

Study	Methods	Dataset	Key findings	Accuracy	Applications	Level
Preethi et al. [9]	SVM, Naive Bayes, KNN	100 images	SVM performs best, pores and wrinkles important features	92%	Dermatology, cosmetics, skincare product development	V
Ding et al. [10]	CNN	1340 images (training), 200 images (testing)	CNN accurately identifies benign pigmented skin lesions	92.5% (sensitivity: 95.8%, specificity: 89.4%)	Reducing workload of dermatologists, improving diagnosis accuracy	IIb
Liao et al. [11]	GAFCM	150 images	GAFCM outperforms traditional methods	92.3%	Real-time skin symptom detection and monitoring	IV
Redmon et al. [12]	YOLO	Benchmark datasets (e.g., PASCAL VOC)	YOLO achieves state-of-the-art results	High average precision and speed	Various real-time object detection applications	Ia
Maknuna et al. [13]	CNN	150 images	CNN classifies scar tissue stages, Scar Index correlates with clinical outcomes	95%	Analyzing and quantifying scar tissue	IV
Wen et al. [14]	CNN (two architectures)	1500 images	Accurate detection and evaluation of acne lesions and severity	93.2% (detection), correlation coefficient: 0.84 (severity)	Clinical practice, remote monitoring, patient assessment	IIIb
Phan et al. [15]	CNN, IoT	2500 images (training), 500 images (testing)	Smart LED therapy device diagnoses acne, monitors treatment	95.6% (F1-score: 0.93)	Remote monitoring, telemedicine	IV
Gallucci et al. [16]	CNN, transfer learning	1000 images (training), 500 images (testing)	Deep learning-based hair counting method	95.3% (precision: 94.5%)	Trichoscopy, hair transplantation	IV
Shih et al. [17]	CNN, transfer learning	1200 images (training), 300 images (testing)	Evaluates effectiveness of vitiligo treatment	92.5%	Clinical trials, treatment evaluation	IV
Ismail et al. [18]	Faster R-CNN with ResNet-50	1000 images (training), 500 images (testing)	Detects acne lesions and wrinkles	95.6% (acne), 92.8% (wrinkles)	Dermatology clinics	IV
Rew et al. [19]	Hybrid segmentation (thresholding and active contour)	200 images (training), 100 images (testing)	Accurate segmentation of skin features	93.5%	Disease diagnosis and treatment	IV
Yap [20]	Review	25 datasets, 23 methods, 17 challenges	Review of facial wrinkles detection and inpainting techniques	N/A	Medical and cosmetic applications, future research directions	V

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generate a detailed image. This high-resolution image is then fed into a classification module that diagnoses the facial skin problem.

The authors evaluated their system on a dataset of 10,000 facial images and achieved a high accuracy rate of 95.2% in diagnosing various skin problems. The system was also able to detect subtle changes in skin texture and color that are indicative of specific skin conditions (Level IV).

Borade et al. [24] proposes a ML-based approach for automatic detection and diagnosis of cosmetic skin types and diseases.

The authors developed a system that uses a combination of CNNs and support vector machines (SVMs) to analyze images of skin samples. The system is trained on a dataset of 1200 images of different skin types and diseases, including acne, eczema, and psoriasis.

The system is divided into two stages: feature extraction and classification. The feature extraction stage uses a CNN to extract features from the input image, and the classification stage uses an SVM to classify the extracted features into one of the predefined skin types or diseases.

The authors evaluated their system on a test dataset of 300 images and achieved an accuracy rate of 92.3%. They also compared their system with other existing methods and found that their system outperformed them in terms of accuracy (Level IIIb).

Vatiwutipong et al. [25] conducted a systematic literature search to identify studies that investigated the use of AI in various aspects of cosmetic dermatology, including skin analysis, diagnosis, treatment planning, and outcome assessment.

The review found that AI has been increasingly used in cosmetic dermatology for tasks such as skin type classification, wrinkle detection, and facial feature analysis. The authors also identified studies that used ML algorithms to predict treatment outcomes, such as the effectiveness of botulinum toxin injections or chemical peels. Additionally, the review highlighted the use of AI-powered systems for automated skin cancer detection and diagnosis.

The authors noted that while AI has shown promising results in cosmetic dermatology, there are still limitations and challenges to be addressed, including the need for large-scale datasets and the potential biases in ML algorithms. The review concludes by highlighting the potential benefits of AI in improving the efficiency and accuracy of cosmetic dermatology services, as well as reducing the risk of human error (Level Ia).

Ito et al. [26] presents a novel approach to scar screening using ML. The authors developed a system that uses a combination of computer vision and ML algorithms to analyze images of scars and predict their severity. The system is trained on a dataset of 1000 images of scars with varying degrees of severity and uses a CNN to extract features from the images.

The system is divided into three stages: feature extraction, feature selection, and classification. The feature extraction stage uses a CNN to extract features from the input image, the feature selection stage selects the most relevant features for classification, and the classification stage uses a random forest algorithm to predict the severity of the scar.

The authors evaluated their system on a test dataset of 500 images and achieved an accuracy rate of 85%. They also compared their system with other existing methods and found that their system outperformed them in terms of accuracy (Level IIc).

Jain et al. [27] presents a novel approach to classify multi-type skin diseases using an optimized deep neural network (OP-DNN) based feature extraction method.

The authors propose a framework that uses OP-DNN to extract relevant features from skin lesion images and then classifies them into different disease categories using a SVM classifier. The OP-DNN is trained on a dataset of 10,000 images of skin lesions with 15 different disease types.

The proposed framework consists of three stages: image preprocessing, feature extraction, and classification. In the first stage, the input images are preprocessed using histogram equalization and contrast stretching techniques. In the second stage, the OP-DNN is used to extract features from the preprocessed images, including texture, shape, and color features. Finally, the extracted features are fed into the SVM classifier to classify the images into their respective disease categories.

The authors evaluated their framework on a test dataset of 2000 images and achieved an accuracy rate of 94%. They also compared their results with other existing methods and found that their framework outperformed them in terms of accuracy and precision (Level IIc).

López-Leyva et al. [28] presents a novel approach to diagnose skin lesions using the Fourier spectral information of images on an additive color model with artificial neural networks.

The authors propose a system that uses the Fourier transform to extract spectral features from images of skin lesions and then classifies them into different disease categories using a multiclass artificial neural network (ANN). The system is trained on a dataset of 1500 images of skin lesions with 12 different disease types.

The proposed system consists of three stages: image preprocessing, feature extraction, and classification. In the first stage, the input images are preprocessed using histogram equalization and contrast stretching techniques. In the second stage, the Fourier transform is applied to extract spectral features from the preprocessed images. Finally, the extracted features are fed into the multiclass ANN to classify the images into their respective disease categories.

The authors evaluated their system on a test dataset of 300 images and achieved an accuracy rate of 93%. They also compared their results with other existing methods and found that their system outperformed them in terms of accuracy and precision (Level IIc).

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Elder et al. [29] highlight the increasing use of AI in various aspects of cosmetic dermatology, including skin analysis, diagnosis, treatment planning, and outcome assessment.

The article discusses the use of ML algorithms to analyze facial images and predict treatment outcomes, such as the effectiveness of botulinum toxin injections or chemical peels. The authors also note the development of AI-powered systems for automated skin cancer detection and diagnosis, as well as the use of AI to analyze patient data to personalize treatment plans.

The authors also discuss the potential benefits of AI in cosmetic dermatology, including improved accuracy and efficiency, reduced radiation exposure from imaging studies, and enhanced patient engagement. However, they also highlight the need for further research to address the limitations and challenges associated with AI adoption, including bias in training datasets and the need for regulatory approval (Level IV).

Huong et al. [30] presents a novel approach to classify skin diseases using customized AlexNet models. The authors propose a DL-based framework that utilizes the AlexNet architecture to classify skin lesions into different disease categories. The proposed framework consists of two stages: pre-processing and classification. In the pre-processing stage, the input images are resized and normalized to improve the quality of the input data. In the classification stage, the pre-processed images are fed into the customized AlexNet models, which are trained using a dataset of 10,000 images of skin lesions with 10 different disease types.

The authors proposed three customized AlexNet models: AlexNet-256, AlexNet-512, and AlexNet-1024, which differ in the number of layers and convolutional layers used. The models were evaluated on a test dataset of 2000 images and achieved an accuracy rate ranging from 92% to 96%, depending on the model used.

The results show that the customized AlexNet models outperformed other existing methods in terms of accuracy and precision. The authors also compared their results with other DL-based approaches and found that their approach achieved better performance (Level IIc).

Yang et al. [31] developed two AI models: a CNN model and a transfer learning-based model using a pre-trained VGG16 network. The models were trained on a dataset of 1200 images of benign, pigmented facial skin lesions and their corresponding histopathological diagnoses. The CNN model was trained from scratch, while the transfer learning-based model used pre-trained weights from the VGG16 network and fine-tuned them on the skin lesion dataset.

The authors evaluated the performance of both models using a test dataset of 240 images and found that the CNN model achieved an accuracy of 92.5%, sensitivity of 93.8%, and specificity of 91.5%. The transfer learning-based model achieved an accuracy of 95.8%, sensitivity of 96.4%, and specificity of 95.2%. The results showed that both models performed well in diagnosing benign, pigmented facial skin lesions (Level IIIb).

Abas et al. [32] developed a system that localizes and classifies acne lesions using a combination of techniques from computer vision and ML.

The system consists of two main components: lesion localization and lesion classification. The lesion localization component uses a CNN to detect the presence of acne lesions in an image and localize them within the image. The lesion classification component uses a SVM to classify the localized lesions into one of three categories: comedones, papules, or pustules.

The authors evaluated their system on a dataset of 1000 acne images and achieved an accuracy of 93% for lesion localization and an accuracy of 85% for lesion classification. They also compared their results with those of human dermatologists and found that their system performed similarly to human experts in terms of accuracy (Level IIc).

3.2 | Diagnosis of Skin Condition

The results are summarized in Table 2.

Bonnekoh et al. [44] explores the latest advancements in dermatology and their potential impact on the field. The authors highlight several innovative technologies and approaches that are transforming the way dermatologists diagnose and treat skin conditions. These include: AI and ML algorithms that can analyze medical images and detect skin cancers earlier and more accurately; Virtual and augmented reality platforms that enable patients to visualize and interact with their skin conditions in a more immersive and engaging way; Personalized medicine approaches that use genetic analysis to tailor treatment plans to an individual's unique genetic profile; Advanced wound care technologies that use nanotechnology, bioactive dressings, and other innovative materials to accelerate wound healing

The authors also discuss the importance of digitalization in dermatology, including the use of mobile apps, telemedicine platforms, and electronic health records to improve patient engagement and care coordination (Level V).

Jeddi et al. [45] presents a novel approach to developing an intelligent diagnostic assistant for complicated skin diseases. The authors propose a decision support system based on the C5.0 algorithm, a type of ML algorithm, to aid dermatologists in diagnosing complex skin conditions.

The authors developed a dataset of 1000 patients with various skin diseases, including psoriasis, vitiligo, and eczema. They used the C5.0 algorithm to analyze the dataset and identify the most relevant clinical features for diagnosis. The resulting model was tested on an independent dataset of 200 patients and achieved an accuracy of 92.5%. The system was able to correctly diagnose 87% of cases with psoriasis, 85% of cases with vitiligo, and 90% of cases with eczema.

The authors also compared their approach with other ML algorithms, such as Naive Bayes and SVMs, and found that C5.0 performed better in terms of accuracy and speed (Level IV).

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TABLE 2 | The table summarized the articles on the estimation of disease severity.

Study	Methods	Dataset	Key findings	Accuracy	Applications	Level
Maroni et al. [33]	CNN	100 images	Proposed system detects, extracts, and counts acne lesions	95%	Acne severity assessment using smartphone images	IIb
Chang et al. [34]	CNN	1200 images	ScalpEye system diagnoses dandruff, seborrheic dermatitis, and alopecia areata	93.5%	Diagnosing scalp health issues	IIb
Man et al. [35]	CNN	1200 SEM images	Detects hair damage, providing early warnings	95.2%	Hair care and damage assessment	IIb
Wang et al. [36]	DCNN	150 CT and MRI scans	Classifies microtia severity into mild, moderate, and severe	92.7%	Assessing microtia severity for treatment planning	IIIb
Wang et al. [37]	Mobile app (AcneScan), ML	1500 images from 300 patients	Classifies acne severity into four categories	85.6%	Mobile acne severity assessment	IV
Seck et al. [38]	3D surface texture analysis, ML	60 subjects (30 healthy, 30 patients)	Analyzes surface texture to classify skin conditions	92.3%	Facial skin condition assessment	IIb
Flament et al. [39]	Automated algorithm	2000 selfie images	Evaluates facial signs such as wrinkles, pores, and skin texture	High agreement with dermatologists	Skincare product development, cosmetic procedures	IIIb
Flament et al. [40]	AI analysis	500,000 selfie images	Investigates differences in facial skin aging between European and Chinese women	N/A	Personalized skincare products	IV
Flament et al. [41]	FSAS algorithm	1000 selfie images	Automated grading system for facial signs, validated against dermatologists' evaluations	High correlation with dermatologists	Dermatological applications	IIIb
Jiang et al. [42]	Automated grading system	Facial images	Grades facial skin structural signs, comparing with expert assessments	High accuracy and reliability	Skincare product development, cosmetic treatments	IIIb
Li et al. [43]	Review of AI-based methods	Various datasets	Overview of AI in grading acne vulgaris, highlighting benefits and future prospects	N/A	Acne treatment and monitoring	V

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Wang et al. [46] developed two ML models: a single-data model that uses a random forest algorithm to analyze a single metagenomic sequencing dataset, and a multi-data model that combines data from multiple datasets using a stacked generalization approach.

The authors evaluated their models on a dataset of 120 acne patients and 120 healthy controls, and found that the single-data model achieved an accuracy of 85.8% and the multi-data model achieved an accuracy of 92.1%. They also compared their results with those of traditional PCR-based methods and found that their ML models performed better.

The authors used a combination of ML algorithms and metagenomic sequencing data to identify biomarkers associated with acne, including genes involved in the pathogenesis of acne, such as propionibacterium acnes and *staphylococcus epidermidis*. They also identified novel biomarkers not previously associated with acne, suggesting potential new therapeutic targets (Level IIIb).

Jansen et al. [47] developed a CNN model using a dataset of 230 images of skin lesions, which were classified as either Bowen's disease or seborrheic keratosis by dermatopathologists.

The CNN model was trained on 160 images and tested on 70 images, achieving an accuracy of 93.1%. The authors also compared their results with those of human dermatologists and found that the CNN model performed similarly to human experts in terms of accuracy. Furthermore, the authors evaluated the CNN model on a separate test set of 30 images and achieved an accuracy of 90%.

The authors used a combination of features from dermatoscopic and histopathological images to train the CNN model, including features such as color, texture, and shape. They also evaluated the performance of the model using several metrics, including precision, recall, and F1-score (Level IIc).

Dubey et al. [48] published in the journal Laser Physics in 2018, presents a novel approach for identifying fungal infections on human scalp using optical coherence tomography (OCT) and ML. The authors developed a system that uses OCT to acquire images of the scalp, which are then analyzed using ML algorithms to identify the presence of fungal infections.

The system was tested on 30 patients with fungal infections and 20 healthy controls, and was able to accurately identify the infections with a sensitivity of 93.3% and specificity of 95%. The authors also compared their results with those of traditional methods, such as microscopy and culture, and found that their system was more accurate and faster.

The authors used a combination of features from the OCT images, including the intensity, texture, and morphology of the skin, to train a SVM model to classify the images as either infected or uninfected. They also evaluated the performance of the system using several metrics, including accuracy, precision, recall, and F1-score (Level 2c).

Alagic et al. [49] developed a system that combines computer vision and ML algorithms to analyze facial skin images and predict the presence of various skin conditions.

The system was trained on a dataset of 500 facial skin images, which were labeled as normal, acne-prone, or rosacea-prone. The authors used a CNN to analyze the images and identify patterns associated with each condition. The system was able to accurately classify the images with an accuracy of 93.6%.

The authors evaluated the performance of the system using several metrics, including precision, recall, and F1-score. They also compared their results with those of human dermatologists, finding that the AI system performed similarly to human experts.

The article highlights the potential benefits of AI-based analysis for improving the accuracy and efficiency of facial skin health assessment. The system could potentially be used in clinical settings to help diagnose and monitor skin conditions, as well as provide personalized recommendations for treatment (Level IV).

Aditya et al. [50] explores the use of ML algorithms for predicting the likelihood of alopecia areata, a chronic autoimmune disease that causes hair loss. The authors analyzed a dataset of 150 patients with alopecia areata and 150 healthy individuals, and used various ML techniques, including decision trees, random forests, and SVMs, to identify patterns and predictors of the disease.

The authors found that the SVM model performed best, achieving an accuracy of 92.3%, sensitivity of 90.7%, and specificity of 93.3%. They also identified several key features that were highly correlated with the presence of alopecia areata, including age, gender, family history, and certain laboratory values.

The study suggests that ML-based models can be useful for predicting the likelihood of alopecia areata, which could potentially aid in early diagnosis and treatment. The authors note that their findings could also be used to develop personalized treatment plans for patients with alopecia areata (Level IIIb).

Goldsberry et al. [51] discusses the VISIA system, a facial analysis software that uses computer vision and AI to evaluate facial skin health and analyze facial features. The authors highlight the potential benefits of this technology for cosmetic practitioners, including its ability to assess skin texture, pores, fine lines, and wrinkles, as well as identify areas of skin damage and recommend personalized treatment plans.

The VISIA system uses a three-step process to analyze the face: first, it captures high-definition images of the face using a specialized camera; second, it analyzes the images using proprietary software; and third, it provides a comprehensive report highlighting areas of concern and recommending treatments. The authors suggest that this technology could be used to improve patient communication and education, enhance treatment planning, and monitor treatment outcomes.

The article is a review of the literature on the VISIA system and its potential applications in cosmetic practice, rather than an original research study. Therefore, it does not provide new data or evidence on the effectiveness of the technology (Level V).

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Liu et al. [52] presents a novel DL-based approach for diagnosing melasma, a common skin condition characterized by hyperpigmentation. The authors developed an intelligent diagnostic model that utilizes multimode image input, including dermoscopy, fundus photography, and confocal microscopy images.

The model was trained on a dataset of 1200 images from 400 patients with melasma and 400 healthy controls. The authors used a CNN to extract features from the images and a SVM to classify the images as melasma-positive or melasma-negative. The model achieved an accuracy of 95.5%, sensitivity of 93.5%, and specificity of 97.5%.

The authors also evaluated the model's performance on an external dataset of 200 images from 100 patients with melasma and 100 healthy controls, achieving an accuracy of 92.5%. They concluded that their intelligent diagnostic model can be a valuable tool for diagnosing melasma and potentially improving treatment outcomes (Level IIb).

Sameera et al. [53] explores the application of DL for real-time detection of various signs of aging. The authors propose a novel approach that uses CNNs to classify images of skin into different aging signs, including wrinkles, fine lines, age spots, and skin dryness.

The authors trained their CNN model using a dataset of 1200 images and evaluated its performance on a separate test set of 300 images. They achieved an accuracy of 92.3%, precision of 91.4%, and recall of 93.5% for detecting various signs of aging. The model was also able to detect multiple signs simultaneously, demonstrating its potential for real-time detection and classification.

The authors suggest that their approach can be used in various applications, including skin cancer detection, dermatology diagnosis, and cosmetic treatment planning. However, they note that their study has limitations, including the use of a small dataset and the need for further evaluation on larger datasets (Level IV).

Huang et al. [54] presents a ML-based approach for evaluating the diagnostic value of non-invasive imaging instruments for rosacea. The authors developed a multi-layer perceptron (MLP) model to analyze a dataset of 240 images of rosacea patients and healthy controls, along with their corresponding clinical data.

The MLP model was trained using a total of 12 features extracted from the images, including morphological and textural features. The authors evaluated the model's performance on a test set of 60 images and achieved an accuracy of 92.3%, sensitivity of 95.7%, and specificity of 90.5%. They also compared the performance of their MLP model with that of three other ML algorithms, including SVMs, random forests, and gradient boosting machines.

The authors concluded that their MLP model outperformed the other algorithms in terms of accuracy and precision, and recommended its use as a valuable tool for assessing the diagnostic value of non-invasive imaging instruments for rosacea. However, they noted that the study had some limitations, including the use of a relatively small dataset and the need for further evaluation on larger datasets (Level IIb).

Boumaraf et al. [55] compares the performance of conventional ML and DL approaches for classifying breast cancer histopathological images. The authors used a dataset of 2000 images with corresponding magnification levels (×40, ×100, and ×200) and compared the accuracy of two ML algorithms (SVMs and random forests) with that of a DL model (CNN).

The results showed that the DL model outperformed the ML algorithms in terms of accuracy, with an average accuracy of 93.5% compared to 83.4% for the ML models. The DL model was also able to provide visual explanations for its predictions, which can aid in understanding the decision-making process. The authors concluded that DL is a promising approach for breast cancer image classification, especially when combined with magnification-dependent features.

The study highlights the importance of considering magnification levels when classifying breast cancer images, as the performance of ML algorithms can be affected by this factor (Level IIb).

3.3 | Estimation of Disease Severity

The results are summarized in Table 3.

Maroni et al. [33] presents a ML-based approach to detect, extract, and count acne lesions from images. The authors propose a novel method that uses CNNs to automatically identify and quantify acne lesions in images captured using a smartphone camera. The proposed system consists of three stages: lesion detection, extraction, and counting.

In the detection stage, the CNN is trained to identify regions of interest (ROI) in the image that correspond to acne lesions. In the extraction stage, the system extracts the detected ROIs and removes noise and other irrelevant features. Finally, in the counting stage, the system counts the number of extracted ROIs to determine the severity of acne.

The authors tested their system using a dataset of 100 images and achieved an accuracy of 95% in detecting acne lesions. They also compared their system with existing methods and demonstrated improved performance (Level IIb).

Chang et al. [34] presents ScalpEye, a DL-based system that uses CNNs to diagnose scalp health issues using images of scalp hair. The system was developed and evaluated on a dataset of 1200 images of scalp hair from healthy individuals and those with various scalp conditions. The authors fine-tuned a pre-trained CNN model on this dataset and achieved an accuracy of 93.5% in diagnosing dandruff, seborrheic dermatitis, and alopecia areata.

The system consists of three main components: hair segmentation, feature extraction, and disease classification. The hair segmentation module separates individual hairs from the

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TABLE 3 | The table provides summary of each study on the diagnosis of skin conditions.

Study	Methods	Dataset	Key findings	Accuracy	Applications	Level
Bonnekoh et al. [44]	AI, ML, VR, AR, Personalized Medicine, Advanced Wound Care	N/A	Discusses innovative technologies transforming dermatology	N/A	Dermatology, wound care, telemedicine	V
Jeddi et al. [45]	C5.0 algorithm (ML)	1000 patients (training), 200 patients (testing)	Intelligent diagnostic assistant for complicated skin diseases	92.5%	Diagnosis of psoriasis, vitiligo, eczema	IV
Wang et al. [46]	Random forest (single-data), Stacked generalization (multi-data)	120 acne patients, 120 healthy controls	Multi-data model outperforms single-data model in acne biomarker identification	92.1% (multi-data), 85.8% (single-data)	Biomarker identification for acne	IIIb
Jansen et al. [47]	CNN	230 images of skin lesions	CNN model for classifying Bowen's disease and seborrheic keratosis	93.1% (training), 90% (test set)	Lesion classification	IIc
Dubey et al. [48]	OCT, SVM	30 patients with fungal infections, 20 healthy controls	OCT and ML system for identifying fungal infections on scalp	Sensitivity: 93.3%, Specificity: 95%	Fungal infection detection	IIc
Alagic et al. [49]	CNN	500 facial skin images	AI system for predicting presence of acne and rosacea	93.6%	Facial skin health assessment	IV
Aditya et al. [50]	SVM	150 alopecia areata patients, 150 healthy individuals	SVM model for predicting likelihood of alopecia areata	92.3% (Sensitivity: 90.7%, Specificity: 93.3%)	Predicting alopecia areata	IIIb
Goldsberry et al. [51]	VISIA system	N/A	Review of facial analysis software for evaluating facial skin health and features	N/A	Cosmetic practice, treatment planning	V
Liu et al. [52]	CNN, SVM	1200 images (training), 200 images (testing)	Multimode image input model for diagnosing melasma	95.5% (training), 92.5% (testing)	Diagnosing melasma	IIb
Sameera et al. [53]	CNN	1200 images (training), 300 images (testing)	Real-time detection of aging signs using deep learning	92.3% (Precision: 91.4%, Recall: 93.5%)	Aging sign detection	IV
Huang et al. [54]	MLP	240 images (training), 60 images (testing)	Evaluating non-invasive imaging for rosacea diagnosis	92.3% (Sensitivity: 95.7%, Specificity: 90.5%)	Rosacea diagnosis	IIb
Boumaraf et al. [55]	SVM, Random forests, CNN	2000 histopathological images	DL model outperforms ML models for classifying breast cancer images, considering magnification levels	93.5% (DL), 83.4% (ML)	Breast cancer image classification	IIb

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image, while the feature extraction module extracts relevant features such as color, texture, and shape from the segmented hairs. The disease classification module uses these features to classify the scalp condition into one of three categories.

The authors conducted a user study with 20 participants who provided feedback on the system's usability and accuracy. The results showed that ScalpEye was able to accurately diagnose scalp conditions and provided users with clear and informative output (Level IIb).

Man et al. [35] presents a CNN based system for detecting hair damage using scanning electron microscopy (SEM) images. Hair damage is a common issue that can be caused by various factors, including chemical treatments, heat styling, and environmental factors. The proposed system aims to identify damaged hair and provide an early warning system for hair care professionals and individuals.

The authors used a dataset of 1200 SEM images of hair samples with different levels of damage, including normal, slightly damaged, and heavily damaged. They trained a CNN model on this dataset and achieved an accuracy of 95.2% in detecting hair damage. The system consists of three main components: image preprocessing, feature extraction, and classification.

The image preprocessing module normalizes the SEM images to enhance their quality and reduce noise. The feature extraction module uses convolutional layers to extract features from the preprocessed images, which are then fed into the classification module. The classification module uses a fully connected layer to classify the hair damage level into one of three categories: normal, slightly damaged, or heavily damaged (Level IIb).

Wang et al. [36] presents a novel approach to assessing the severity of microtia, a congenital deformity characterized by underdeveloped or absent outer ear. The authors used a deep convolutional neural network (DCNN) to analyze computed tomography (CT) scans and magnetic resonance imaging (MRI) of patients with microtia and compared the results with those from traditional radiological assessments.

The DCNN was trained using a dataset of 150 CT and MRI scans from patients with microtia, and the model was able to accurately classify the severity of microtia into three categories: mild, moderate, and severe. The authors evaluated the performance of the DCNN using several metrics, including accuracy, sensitivity, and specificity. The results showed that the DCNN achieved high accuracy (92.7%) and sensitivity (91.8%) in classifying microtia severity, outperforming traditional radiological assessments.

The study highlights the potential of AI in improving the assessment of microtia severity, which can have significant implications for patient care and treatment planning. Current methods for assessing microtia severity are often subjective and rely on expert interpretation of imaging studies, which can lead to interobserver variability. In contrast, AI-based approaches can provide objective and reproducible assessments, potentially reducing errors and improving patient outcomes (Level IIIb).

Wang et al. [37] presents a novel mobile application (app) for assessing facial acne severity using a ML-based algorithm. The app, called "AcneScan," uses a smartphone's camera to capture images of the face and then analyzes the images to evaluate the severity of acne based on factors such as lesion size, distribution, and inflammation.

The authors developed and tested the AcneScan app using a dataset of 1500 images from 300 patients with acne, and compared the results with those from traditional clinical assessments performed by dermatologists. The results showed that the AcneScan app was able to accurately classify acne severity into four categories: mild, moderate, severe, and very severe, with an accuracy rate of 85.6%. The app was also able to identify specific features associated with acne severity, such as comedone size and inflammatory lesion count.

The study highlights the potential of mobile technology in providing a convenient and accessible tool for assessing facial acne severity. Acne is a common skin condition that can have a significant impact on quality of life, and accurate assessment is essential for effective treatment. However, current methods for assessing acne severity are often subjective and may vary between healthcare providers. The AcneScan app provides an objective and standardized approach to assessing acne severity, which could potentially improve patient care (Level IV).

Seck et al. [38] presents a novel approach to assessing facial skin condition using three-dimensional (3D) surface texture analysis. The authors developed a method to analyze the surface texture of the skin using high-resolution normal fields (HRNFs), which are maps of the skin's surface topography. The HRNFs were obtained using a commercial instrument called the Xystrom Skin Analyzer.

The study included 30 healthy subjects and 30 patients with various skin conditions, including acne, psoriasis, and atopic dermatitis. The HRNFs were analyzed using a software program that extracted features such as roughness, waviness, and skewness. The authors then used ML algorithms to classify the skin conditions based on the extracted features.

The results showed that the 3D surface texture analysis was able to accurately classify the skin conditions with an accuracy rate of 92.3%. The study found that the roughness feature was most effective in distinguishing between healthy skin and skin conditions, while waviness was most effective in distinguishing between different skin conditions.

The study highlights the potential of 3D surface texture analysis as a non-invasive and objective method for assessing facial skin condition. Current methods for assessing skin condition are often subjective and may vary between healthcare providers. This novel approach could potentially provide a standardized and accurate way to assess skin condition, which could improve patient care and treatment outcomes (Level IIb).

Flament et al. [39] presents a study on the accuracy and clinical relevance of an automated algorithm-based analysis of facial signs from selfie images in women of different ages, ancestries, and phototypes. The study aimed to evaluate the performance

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of the algorithm in assessing facial signs such as wrinkles, pores, and skin texture.

The authors analyzed a dataset of 2000 selfie images from women in the United States with varying ages (18–65 years), ancestries (Caucasian, African American, Asian, Hispanic), and phototypes (I–VI). The images were analyzed using an automated algorithm that assessed facial signs and assigned a score for each sign. The authors compared the algorithm's scores with those obtained from dermatologists' clinical evaluations.

The results showed that the algorithm accurately detected facial signs with a high degree of agreement with dermatologists' evaluations. The study found that the algorithm performed well across different age groups, ancestries, and phototypes. The authors concluded that the algorithm has potential clinical relevance in providing a standardized and objective assessment of facial signs, which could be useful in various dermatological applications such as skincare product development and cosmetic procedures (Level IIIb).

Flament et al. [40] presents a study on the facial skin ageing process in European and Chinese women using AI analysis. The study aimed to investigate the differences in facial skin ageing between the two populations.

The authors analyzed a dataset of over 500,000 selfie images from European and Chinese women, aged 18–65 years. The images were analyzed using an AI algorithm that assessed facial signs such as wrinkles, pores, and skin texture. The authors compared the results between the two populations and identified striking differences in the facial skin ageing process.

The study found that European women experienced a more gradual and uniform ageing process, whereas Chinese women experienced a more rapid and heterogeneous ageing process. The results showed that Chinese women had more prominent wrinkles, deeper pores, and more uneven skin texture than European women. The authors also found that Asian-specific genetic and environmental factors contributed to these differences.

The study highlights the potential of AI analysis in revealing insights into the facial skin ageing process and identifying population-specific differences. The findings have implications for the development of personalized skincare products and treatments tailored to specific populations (Level IV).

Flament et al. [41] presents a study on the development and validation of an automated grading system for facial signs in selfie images. The system, called the Facial Skin Analysis System (FSAS), aims to provide a complete and objective assessment of facial signs such as wrinkles, pores, and skin texture.

The study included 1000 selfie images from Asian women with three different ethnic origins (Korean, Chinese, and Japanese) and ages ranging from 18 to 65 years. The images were analyzed using the FSAS algorithm, which assigns a score for each facial sign. The authors validated the system by comparing the scores with dermatologists' evaluations.

The results showed that the FSAS system demonstrated high reliability and accuracy in assessing facial signs across different ethnic groups and age ranges. The system performed well in detecting wrinkles, pores, and skin texture, with high correlation coefficients with dermatologists' evaluations. The authors also found that the system was able to detect subtle differences in facial signs between the three ethnic groups.

The study highlights the potential of the FSAS system in providing a standardized and objective assessment of facial signs, which can be used in various dermatological applications such as skincare product development and cosmetic procedures (Level IIIb).

Jiang et al. [42] introduces a novel automated procedure for grading facial skin structural signs without the need for human assessment. The system aims to objectively assess specific facial signs related to skin ageing, such as wrinkles, pores, and skin texture.

The study involved comparing the automated grading system's assessments with evaluations by experts using referential atlases of skin ageing. The researchers analyzed a dataset of facial images and evaluated the accuracy and reliability of the automated grading system in detecting and grading facial structural signs.

The results showed that the new automated procedure demonstrated high accuracy and reliability in grading facial skin structural signs compared to assessments by experts. The system was able to effectively detect and grade wrinkles, pores, and skin texture, showing promise as an objective tool for skin ageing assessment.

The study highlights the potential of automated systems in providing consistent and reliable assessments of facial skin ageing, reducing the subjectivity associated with human evaluations. The new procedure could have implications for skincare product development, cosmetic treatments, and dermatological research (Level IIIb).

Li et al. [43] provides an overview of the current state and future prospects of AI in grading acne vulgaris. Acne vulgaris is a common skin condition characterized by comedones, papules, pustules, nodules, and cysts, and grading its severity is crucial for treatment planning.

The authors review the current AI-based methods for grading acne vulgaris, including DL-based approaches that use CNNs to analyze images of the skin. These methods have shown promising results in accurately assessing acne severity, outperforming traditional human-based grading systems.

The article highlights the advantages of AI-based grading systems, including objectivity, speed, and scalability. AI can process large amounts of data quickly and accurately, reducing the risk of human error and bias. Additionally, AI-based systems can be trained on diverse datasets, allowing for more comprehensive and robust assessments.

The authors also discuss the potential applications of AI-based grading systems in acne treatment, such as personalizing

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treatment plans based on individual patient characteristics and monitoring treatment outcomes. They also touch on the need for further research to improve the accuracy and generalizability of AI-based grading systems (Level V).

3.4 | Detection and Localization of the Skin Conditions

The results are summarized in Table 4.

Preethi et al. [9] explores the application of ML techniques to analyze human skin texture. The authors use a dataset of 100 images of human skin, collected from various sources, to develop a ML model that can classify skin texture based on its features such as pores, wrinkles, and fine lines.

The authors employ three ML algorithms: SVMs, Naive Bayes, and K-Nearest Neighbors (KNN) to analyze the skin texture images. The results show that the SVM algorithm performs best, achieving an accuracy of 92% in classifying skin textures into four categories: normal, dry, oily, and combination skin. The authors also analyze the features that contribute most to the classification process and find that pores and wrinkles are the most important features for distinguishing between different skin textures.

The article also discusses the potential applications of this ML approach in various fields such as dermatology, cosmetics, and skincare product development (Level V).

Ding et al. [10] presents a DL-based approach to automatically identify benign pigmented skin lesions from clinical images. The authors develop a CNN model that uses a dataset of 1340 clinical images of skin lesions to train and validate the model.

The model was trained to classify skin lesions as either benign or malignant, and was evaluated using a test dataset of 200 images. The results showed that the CNN model achieved an accuracy of 92.5%, with a sensitivity of 95.8% and a specificity of 89.4%. The model was also able to accurately identify benign pigmented skin lesions, such as seborrheic keratoses and solar lentigines, and differentiate them from malignant melanomas.

The authors highlight the potential benefits of this approach, including the ability to reduce the workload of dermatologists and improve the accuracy of diagnoses. They also discuss the limitations of the study, including the need for larger and more diverse datasets to improve the generalizability of the model (Level IIb).

Liao et al. [11] presents an optimization-based approach for detecting face skin symptoms using a novel technology called "Genetic Algorithm-based Fuzzy C-Means (GAFCM)". The authors propose a method that uses a combination of genetic algorithms and fuzzy C-means clustering to analyze images of the skin and identify symptoms such as acne, rosacea, and hyperpigmentation.

The GAFCM algorithm is used to segment the skin images into different regions, and then a set of features is extracted from each region to characterize the skin symptoms. The authors use a dataset of 150 images to train and validate the model, and report an accuracy of 92.3% in detecting face skin symptoms.

The authors claim that the proposed method outperforms traditional methods such as SVMs and KNNs in terms of accuracy and robustness. They also suggest that the GAFCM algorithm can be used for real-time skin symptom detection and monitoring (Level IV).

Redmon et al. [12] presents a real-time object detection algorithm called YOLO (You Only Look Once) that detects objects in images with high accuracy and speed. The authors propose a novel approach that simultaneously detects and classifies objects in a single pass over the image, eliminating the need for region proposal networks (RPNs) and subsequent bounding box regression.

The YOLO algorithm is based on a single neural network that predicts object bounding boxes and class probabilities directly from full images, without requiring intermediate feature extraction or object proposal generation. The authors demonstrate the effectiveness of YOLO on several benchmark datasets, including the PASCAL VOC dataset, achieving state-of-the-art results in terms of average precision and speed.

YOLO's performance is attributed to its ability to learn hierarchical feature representations and predict object locations directly from image pixels. The algorithm's efficiency is further enhanced by its ability to process entire images at once, rather than individual regions or patches (Level Ia).

Maknuna et al. [13] presents a novel approach for analyzing and quantifying scar tissue using ML techniques. The authors propose a method that uses DL algorithms to automatically analyze histopathological images of scar tissue and extract relevant features, such as collagen density, fiber orientation, and collagen type.

The authors used a dataset of 150 images of scar tissue samples and trained a CNN to classify the samples into different stages of scar maturation. The CNN was able to accurately classify the samples with an accuracy of 95%, outperforming traditional manual methods.

The authors also developed a new metric, called the Scar Index, which combines multiple features to quantify the extent of scar tissue. The Scar Index was found to be highly correlated with clinical outcomes, such as patient recovery rates and quality of life (Level IV).

Wen et al. [14] presents a novel approach for detecting and evaluating acne using interpretable CNN models. The authors propose two CNN architectures, one for detecting acne lesions and another for evaluating acne severity, using a dataset of 1500 images of facial skin.

The acne detection model uses a combination of convolutional and recurrent layers to identify acne lesions, achieving an accuracy of 93.2% and an F1-score of 0.88. The acne severity evaluation model uses a CNN to predict the severity of acne based on features extracted from the lesion images, achieving a

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TABLE 4 | The table summarizes each study on the classification of skin conditions.

Study	Methods	Dataset	Key findings	Accuracy	Applications	Level
Molton et al. [21]	3D analysis, computer vision, ML	30 patients	3D system for facial rejuvenation assessment, high accuracy in detecting changes post-surgery	92%	Cosmetic medicine, facial rejuvenation	IIb
Jeong et al. [22]	CNN (AI-ScalpGrader)	1200 images (training), 300 images (testing)	AI system for diagnosing and classifying scalp conditions	94.3%	Scalp health diagnosis	IV
Kim et al. [23]	Enhanced Mask R-CNN, SRGAN	10,000 facial images	Two-stage system for diagnosing facial skin problems, detects subtle changes in skin texture and color	95.2%	Facial skin problem diagnosis	IV
Borade et al. [24]	CNN, SVM	1200 images (training), 300 images (testing)	System for automatic detection and diagnosis of cosmetic skin types and diseases	92.3%	Cosmetic dermatology	IIIb
Vatiwutipong et al. [25]	Systematic review	N/A	Review of AI applications in cosmetic dermatology, highlights benefits and challenges	N/A	Cosmetic dermatology, skin cancer detection	Ia
Ito et al. [26]	CNN, random forest	1000 images (training), 500 images (testing)	System for scar severity prediction	85%	Scar screening	IIc
Jain et al. [27]	OP-DNN, SVM	10,000 images (training), 2000 images (testing)	Framework for classifying multi-type skin diseases	94%	Skin disease classification	IIc
López-Leyva et al. [28]	Fourier transform, ANN	1500 images (training), 300 images (testing)	System for diagnosing skin lesions using Fourier spectral information	93%	Skin lesion diagnosis	IIc
Elder et al. [29]	AI systems	N/A	Review of AI use in cosmetic dermatology, highlights benefits and challenges	N/A	Cosmetic dermatology	IV
Huong et al. [30]	Customized AlexNet models	10,000 images (training), 2000 images (testing)	Deep learning-based framework for classifying skin diseases	92%–96%	Skin disease classification	IIc
Yang et al. [31]	CNN, VGG16 (transfer learning)	1200 images (training), 240 images (testing)	AI models for diagnosing benign pigmented facial skin lesions	92.5% (CNN), 95.8% (VGG16)	Facial skin lesion diagnosis	IIIb
Abas et al. [32]	CNN, SVM	1000 acne images	System for localizing and classifying acne lesions	93% (localization), 85% (classification)	Acne lesion classification	IIc

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correlation coefficient of 0.84 with dermatologist-assessed scores.

The authors claim that their models are interpretable, allowing for visualizing the features used for prediction and providing insights into the decision-making process. They also demonstrate the potential applications of their models in clinical practice, including remote monitoring and patient assessment (Level IIIb.)

Phan et al. [15] presents a novel approach to diagnosing acne vulgaris using a smart LED therapy device that integrates DL and the Internet of Things (IoT). The device uses a CNN to analyze images of the skin and diagnose acne, and then uses IoT technology to remotely monitor and control the therapy process.

The authors trained a CNN model using a dataset of 2500 images of acne-affected skin and tested its performance using an additional 500 images. The model achieved an accuracy of 95.6% and an F1-score of 0.93 in diagnosing acne vulgaris. The device was also evaluated in a clinical trial involving 30 patients, and the results showed that it was able to accurately diagnose acne severity and monitor treatment progress.

The authors claim that their device has several advantages over traditional methods, including increased accuracy, reduced cost, and improved patient convenience. They also highlight the potential for remote monitoring and telemedicine applications (Level IV).

Gallucci et al. [16] presents a novel approach to hair counting using DL techniques. The authors aim to develop an accurate and efficient method for hair count, which is a crucial step in various medical applications, such as trichoscopy and hair transplantation.

The authors propose a DL-based approach using CNNs and transfer learning to classify hair follicles into three categories: active, inactive, and missing. They train the model using a dataset of 1000 images of scalp images and evaluate its performance using an additional 500 images.

The results show that the proposed approach achieves an accuracy of 95.3% and a precision of 94.5% in detecting hair follicles, outperforming traditional methods such as manual counting and thresholding. The authors also demonstrate the potential of their approach for real-world applications by applying it to images from a trichoscopy system (Level IV).

Shih et al. [17] presents a novel approach to evaluating the effectiveness of vitiligo treatment using a weakly supervised ML method. Vitiligo is a chronic autoimmune skin condition characterized by the loss of pigment-producing melanocytes, and accurate evaluation of treatment outcomes is crucial for optimal patient care.

The authors develop a DL-based system that uses a combination of CNNs and transfer learning to classify images of patients' skin before and after treatment. The system is trained on a dataset of 1200 images and evaluated using an additional 300 images.

The results show that the proposed system achieves an accuracy of 92.5% in distinguishing between treatment responders and non-responders, outperforming traditional methods such as visual assessment by dermatologists. The authors also demonstrate the potential of their system for real-world applications by applying it to images from clinical trials (Level IV).

Ismail et al. [18] presents a novel approach to detecting acne lesions and wrinkles using a DL-based method. Acne is a common skin condition that can cause significant psychological distress, while wrinkles are a natural part of the aging process. Accurate detection of these skin conditions is crucial for effective treatment and management.

The authors propose using Faster R-CNN (Region-based Convolutional Neural Networks) with a ResNet-50 architecture to detect acne lesions and wrinkles. The system is trained on a dataset of 1000 images of skin samples and evaluated using an additional 500 images.

The results show that the proposed system achieves an accuracy of 95.6% in detecting acne lesions and 92.8% in detecting wrinkles, outperforming traditional methods such as manual annotation and thresholding. The authors also demonstrate the potential of their system for real-world applications by applying it to images from dermatology clinics (Level IV).

Rew et al. [19] presents a novel approach to segmenting skin features from dermoscopy images using a hybrid method. Dermoscopy is a non-invasive imaging technique used to examine the skin for early detection of skin cancers and other skin conditions. Accurate segmentation of skin features is crucial for disease diagnosis and treatment.

The authors propose a hybrid segmentation scheme that combines two approaches: thresholding and active contour model-based segmentation. The system is trained on a dataset of 200 dermoscopy images and evaluated using an additional 100 images.

The results show that the proposed system achieves an accuracy of 93.5% in segmenting skin features, outperforming traditional methods such as thresholding alone. The authors also demonstrate the potential of their system for real-world applications by applying it to images from clinical datasets (Level IV).

Yap [20] provides a comprehensive review of the current state-of-the-art in facial wrinkles detection and inpainting techniques. Facial wrinkles are a common concern in both medical and cosmetic applications, and detecting them accurately is crucial for skin analysis and treatment.

The authors survey 25 datasets, 23 methods, and 17 challenges related to facial wrinkles detection and inpainting. They categorize the methods into three main categories: feature-based, DL-based, and physics-based methods. The review highlights the limitations of each category and identifies the most promising approaches.

The authors also discuss the challenges associated with facial wrinkles detection, including data quality, variability in facial expressions, and image quality. They propose future directions for research in this area, including the development of more

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robust datasets, the integration of multiple modalities (e.g., RGB and depth images), and the exploration of transfer learning techniques (Level V).

4 | Discussion

The integration of AI and ML in aesthetic medicine has been gaining significant attention in recent years. One of the key findings of the review is that AI and ML have been increasingly used in various aspects of aesthetic medicine, including skin analysis, diagnosis, treatment planning, and outcome assessment.

The review also highlights the potential limitations and challenges associated with AI adoption in aesthetic medicine. One of the main challenges is the need for high-quality training data to develop accurate AI algorithms. Another challenge is the need for regulatory oversight to ensure that AI-powered systems are used safely and effectively. Additionally, there are concerns about issues such as algorithmic bias and data privacy.

The implications of this review are significant for both clinicians and patients. Clinicians should be aware of the potential benefits and limitations of AI-powered systems in aesthetic medicine and consider incorporating these systems into their practice. Patients should also be aware of the potential benefits of AI-powered systems and be willing to participate in clinical trials testing these systems.

The future directions for AI and ML in aesthetic medicine are numerous. One potential direction is the development of more advanced algorithms that can integrate multiple sources of data to improve diagnostic accuracy. For example, algorithms that integrate clinical data with imaging data could provide more accurate diagnoses than those that rely solely on clinical data.

Another potential direction is the development of more user-friendly interfaces for patients to interact with AI-powered systems. Currently, many AI-powered systems require a high level of technical expertize to use effectively. Developing user-friendly interfaces could make it easier for patients to use these systems and increase their adoption.

There is also a need for further research on the ethics of AI and ML in aesthetic medicine. As these technologies become increasingly prevalent in clinical practice, it is essential that we address issues such as algorithmic bias, data privacy, and patient autonomy [56–58].

5 | Conclusion

The integration of AI and ML in aesthetic medicine has revolutionized the way dermatologists and healthcare professionals diagnose, classify, localize, detect, and estimate the severity of skin conditions. This review has highlighted the significant benefits of AI and ML in improving diagnostic accuracy, enhancing patient outcomes, and reducing healthcare costs.

The results of the review have shown that AI-powered systems have been increasingly used in various aspects of aesthetic medicine, including skin analysis, diagnosis, treatment planning, and outcome assessment. The studies included in this review have demonstrated high levels of accuracy and precision in detecting and diagnosing skin conditions such as skin cancer, acne, psoriasis, and seborrheic dermatitis.

Moreover, AI-powered systems have been shown to improve treatment outcomes by providing personalized treatment plans and reducing the risk of human error. For instance, AI-powered systems have been used to develop personalized skincare routines for patients with acne-prone skin, resulting in improved acne severity scores.

However, the review has also highlighted the limitations and challenges associated with AI adoption in aesthetic medicine. The need for high-quality training data, regulatory oversight, and addressing algorithmic bias and data privacy are some of the key challenges that need to be addressed.

As AI-powered systems continue to play a more prominent role in aesthetic medicine, it is essential that clinicians stay up-to-date with the latest advancements and guidelines for using these technologies safely and effectively. Patients should also be aware of the potential benefits and limitations of AI-powered systems and be willing to participate in clinical trials testing these systems.

The integration of AI and ML in aesthetic medicine is a promising area of research that has significant potential benefits for patients and clinicians alike. However, further research is needed to address the limitations and challenges associated with AI adoption in aesthetic medicine.

Author Contributions

Conceptualization: Lisa Kwin Wah Chan, Kar Wai Alvin Lee, Lee Cheuk Hung. Writing—Original Draft Preparation: Kar Wai Alvin Lee, Lisa Kwin Wah Chan. Writing—Review & Editing: Lisa Kwin Wah Chan, Kar Wai Alvin Lee, Jovian Wan, Kyu-Ho Yi. Supervision: Kyu-Ho Yi, Jair Mauricio Cerón Bohórquez, Diala Haykal. All authors have reviewed and approved the article for submission.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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